JONATHAN KUNIARDI

AI & BACKEND ENGINEERING INTERN

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PROFESSIONAL SUMMARY

Computer Science student at BINUS University (Intelligent Systems, GPA 3.72/4.00) who builds voice-driven AI systems in Python. Built a real-time AI VTuber pipeline and measured where its latency goes, and own the backend of a five-person ESP32 voice-assistant project. Available for a full-time internship in semesters 6-7 (BINUS Enrichment Program).

EDUCATION

Bachelor of Computer Science | Intelligent Systems | BINUS University, Jakarta — Sep 2024 - Present
GPA 3.72 / 4.00 · Semester 5 · Expected graduation 2028

Relevant Coursework: Machine Learning, Natural Language Processing, Artificial Intelligence, Computer Vision, Data Structures, Software Engineering, Embedded Systems; in progress: Speech Recognition, Deep Learning

SMA Permai | Natural Sciences (IPA), Jakarta — 2021 - 2024

PROJECTS

Mika IoT: Voice Assistant Satellite | Python, asyncio, WebSocket, faster-whisper, Gemini API, mDNS, pytest — 2026 - Present
github.com/FJGB22/mika-voice-assistant · group project (5 members), Embedded Systems course · backend owner

- Built the server an ESP32-S3 device talks to: it receives 20 ms audio frames over WebSocket, detects the end of speech on the server, transcribes locally with faster-whisper, and sends back Gemini's reply, keeping a separate conversation memory per connection.
- Made the server discoverable as mika.local over mDNS, so the device finds it on any Wi-Fi without reflashing, with a LAN-IP fallback for networks that block multicast.
- Kept protocol rules apart from the network layer and the LLM vendor behind one function, so a new transport or provider means rewriting one module.
- Developed each stage test-first against an 80+ test suite that runs without a microphone, network, or API key; a pre-commit hook blocks any commit that fails lint, format, or tests.

Mika: AI VTuber | Python, asyncio, Gemini API, faster-whisper, edge-tts — 2026 - Present
github.com/FJGB22/ai-vtuber · individual, self-initiated

- Designed a real-time voice pipeline of 6 decoupled modules (speech recognition, turn-taking, streaming LLM, interruptible speech output, avatar control, configuration), mixing a microphone thread with asyncio.
- Implemented sentence-level streaming so speech starts before the reply is finished, and an audio queue that stops when the user speaks.
- Built a measurement bench: first audio arrives at 4.4 s (median of 3 runs), and the current endpoint returns each reply in one piece, so the next target is time-to-first-sentence and TTS start-up, not a bigger model.
- Tested the persona under pressure: 5/5 classic prompt-injection attacks deflected in character, and persona rules held in 27/27 turns across 3 drift runs.

ORGANIZATIONAL EXPERIENCE

Binusian Gaming (BG) | Regional Secretary, Kemanggisan | Jakarta — Jan 2026 - Present

- Own the formal documentation of a region with ~30 committee members and 100+ members per intake: meeting minutes, event proposals, accountability reports (LPJ), and MoU/MoA.
- Took on the secretary workload from late 2025, months before the formal appointment, after a year as an activist (since Nov 2024).

BINUS New Student Expo | Booth Lead, Binusian Gaming | Jakarta — 2025 - 2026

- Led the organization's booth two years running: set the concept, recruited and briefed a crew of under 10, and reached 150+ incoming students at each expo.

BINUS Student Mentoring | Academic Mentor | Jakarta — Sep 2025 - Jan 2026

- Mentored a group of 5 junior students for a full semester while in my own third semester, giving on-demand help across most of their courses instead of following a fixed syllabus.

SKILLS

Programming: Python (primary); Java, TypeScript, HTML, CSS (foundational)

AI & Speech: LLM integration (Gemini API, OpenAI-compatible clients), prompt & persona design, speech-to-text (faster-whisper), text-to-speech (edge-tts), voice activity detection

Backend: asyncio, threading, WebSocket servers, JSON protocol design, mDNS service discovery

Tools & Practices: Git & GitHub, VS Code, pytest, Ruff, pre-commit hooks, benchmarking, AI-assisted development, Figma, Canva

Soft Skills: Formal & technical documentation, event leadership, mentoring

Spoken Languages: Indonesian (native), English (intermediate)